

EVALUATING MARKET RISK MODELS

GROUP 17-05

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1. Definition of VaR & Descriptive Statistics

A SPECIAL CASE OF MARKET RISK: THEORETICAL & DESCRIPTIVE ANALYSIS

MARKET RISK

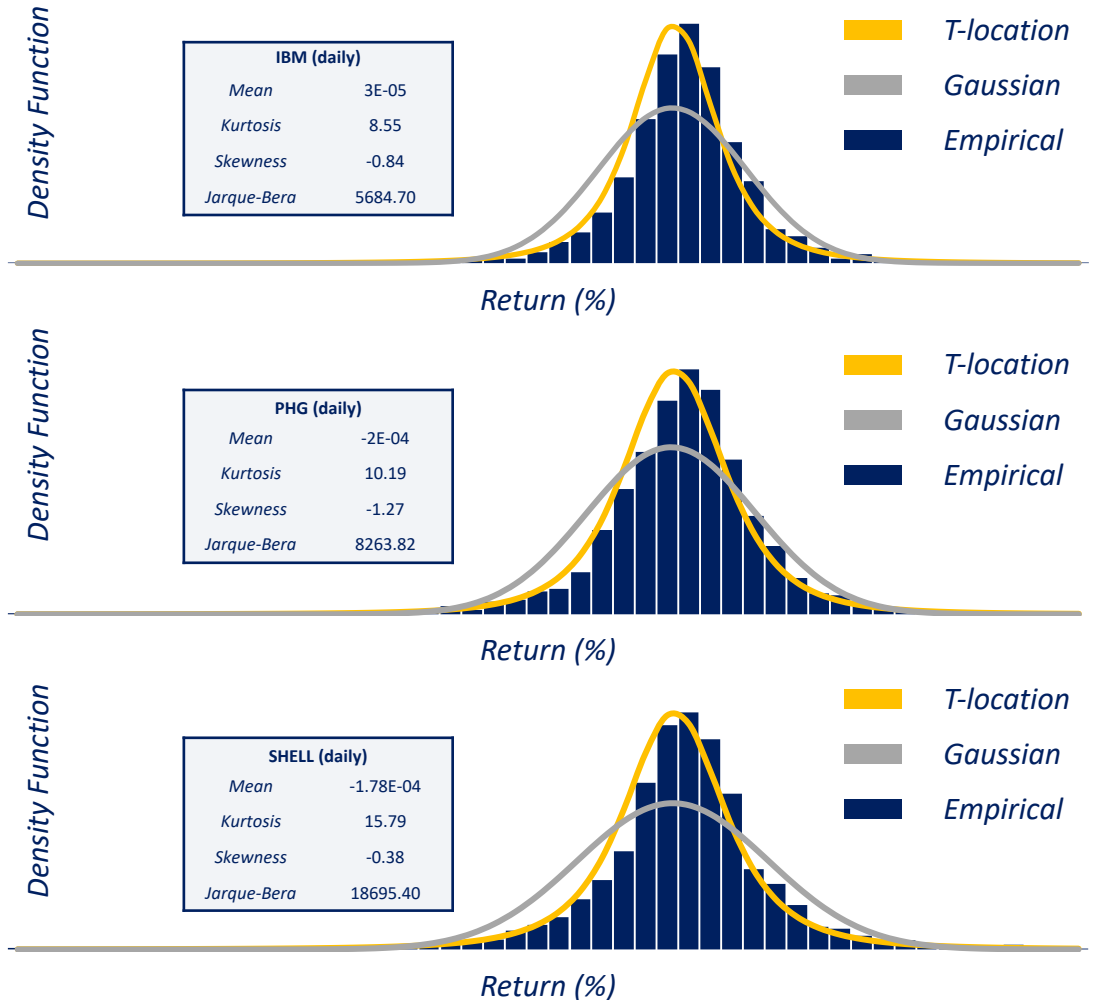
Market risk is the risk of changes in the market value of an instrument or portfolio of financial instruments connected with **unexpected changes in market conditions**: it concerns all the financial assets and liabilities.

The case under analysis relates to a precise specification of market risk: **equity risk** → the sensitivity of the market value of a position to equity market performance.

VALUE AT RISK

Capturing the market risk through the Value at Risk metrics answers the question of *what is the maximum loss which could be suffered over a certain time horizon and with a defined confidence level α* so that the probability of the loss exceeding the VaR is:

$$P(L > VaR) = 1 - \alpha$$



2. Parametric VaR: Analysis

APPROACH TO ESTIMATION

To estimate the parametric VaR the following steps were taken:

1. Assumptions of Gaussian distribution for log returns $r_j \sim N(\mu; \sigma)$: mean prudentially set to 0 ($\mu_j = 0$) and volatility estimated by RiskMetrics (1996) with a decay factor $\lambda = 0.94$

2. 95% confidence level, one day time horizon.

3. Computation of univariate VaR as: $VaR_{uni} = V_j(e^{-\sigma \cdot Z_\alpha} - 1)$

4. Said $\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$ and $\mathbf{C} = \begin{bmatrix} 1 & \rho_{12} & \rho_{13} \\ \rho_{21} & 1 & \rho_{23} \\ \rho_{31} & \rho_{32} & 1 \end{bmatrix}$ respectively the vector of the univariate VaRs and the correlation matrix of returns estimated via EWMA, the multivariate VaR of the equally weighted portfolio is:

$$VaR_{multi} = \sqrt{\mathbf{v}' * \mathbf{C} * \mathbf{v}}$$

Advantages:

- ✓ *Easy to perform* with respect to calculation intensive simulation approaches
- ✓ *Low amount of historical data* required with respect to non-parametric approaches

Disadvantage:

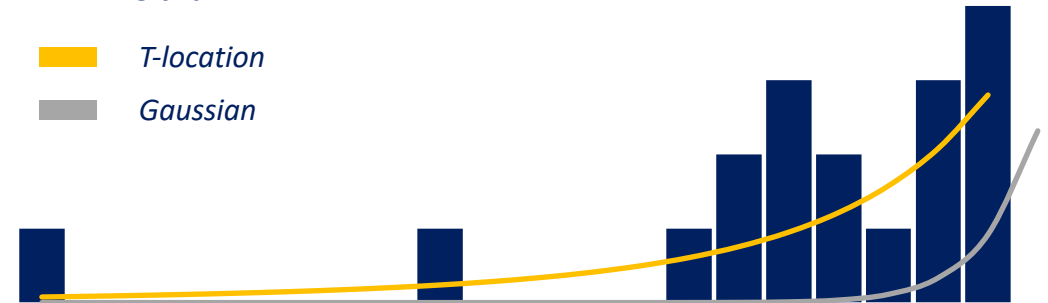
- ✗ *Specific distribution assumed for the random variable may not be confirmed by the real behavior of returns*

DANGEROUS UNDERESTIMATION

IBM - Left Tail

■ T-location

■ Gaussian



- The **Gaussian** fitted with Maximum Likelihood shows a **large degree of underestimation** of extreme events on the left tail.
- T-location scale is estimated manually with Maximum Likelihood and ME algorithm (Excel spreadsheets Digamma, MLE_ME_IBM, ...)
- The **T-location distribution** could provide a **safer approach** to VaR estimation even if extreme events are, also in this case, underestimated.
- PDF of a t-location is fully described by 3 parameters *i.e.* (v, σ, μ) :

$$\frac{\Gamma\left(\frac{v+1}{2}\right)}{\sigma\sqrt{v\pi}\Gamma\left(\frac{v}{2}\right)} \left[\frac{v + \left(\frac{x-\mu}{\sigma}\right)^2}{v} \right]^{-\left(\frac{v+1}{2}\right)}$$

3. Parametric VaR: Estimation & Results

VOLATILITIES ESTIMATION

Volatility tends to happen in **clusters**, so the necessity to model volatility is crucial in the estimation of VaR. Two approaches are possible:

- 1. **GARCH** estimation
- 2. **EWMA** estimation

By choosing the EWMA approach as in RiskMetrics (1996), even if it is a *biased* estimator, many **advantages** arise:

- It is more responsive to recent shocks
- It avoids echo effect
- Upward biased: non-centered second moment → more conservative but still a good approximation.

We decided to use a EWMA instead of a GARCH since the GARCH can be considered too discretionary, as the number of lags is not predetermined. The decay factor used to estimate volatility is $\lambda = 0.94$, generally applied for daily returns.

	IBM	PHG	SHEL
Var Ret	0.016%	0.072%	0.018%
Vol Ret	1.27%	2.68%	1.34%

CORRELATIONS AND RESULTS

Many ways to measure correlations are possible, two example are:

(1) Using the whole dataset without smoothing

(2) Using a smoothed estimate $\rho_{j;k,t} = \frac{\sum_{i=0}^n \lambda^i r_{j,t-i} r_{k,t-i}}{\sigma_{j,t} \sigma_{k,t}}$

EWMA estimate for correlations is more sensitive to recent co-movements of returns.

DAILY SMOOTHED			
Correlation matrix	IBM	PHG	SHEL
IBM	1	0.3	0.06
PHG	0.3	1	-0.07
SHEL	0.06	-0.07	1

DAILY HISTORICAL			
Correlation matrix	IBM	PHG	SHEL
IBM	1	0.3	0.42
PHG	0.3	1	0.26
SHEL	0.42	0.26	1

Possible reason of discrepancies between Historical and Smoothed: different path of Energy Sector (Shell) vs Tech Industry (IBM & Philips) in recent years. Historical correlations *significantly higher* than smoothed ones lead to more **conservative** VaR ^[1].

Multivariate Parametric EW portfolio VaR		Correlations	
		Historical	Smoothed
Distribution	Gaussian	6.45	5.67

[1] T-location scale parametric VaRs are reported only in the Excel file (Sheet T_loc_VaR) since the parametric VaR we finally decide to present is the Gaussian one.

5. Hybrid VaR: Analysis

OVERCOMING THE LIMITATIONS OF THE PARAMETRIC VaR: THE HYBRID APPROACH

The hybrid approach of Boudoukh, Richardson and Whitelaw (1998) combines the merits of the historical simulation approach (no assumptions on the distribution of stock returns) with the use of exponentially decreasing weights given to past observations, in order to give to recent observations more importance.

The **assignment of weights** makes the time of the shock matters, overcoming the limitation of the historical approach, where an extremely negative return, even if happened at different times and market conditions, has the same impact as a new one.

In the sample analyzed, the lowest returns happened on the 9th and on the 12th of March 2020: both observations share a negligible weight in the hybrid approach.

Given a predefined confidence level, we consider (1) the closest observation where the level α is almost reached or (2) we interpolate the two adjacent values of the theoretical percentile.

T-i	Weight	Return	Cum weight
-742	0.0000%	-13.5542%	0.0000%
-745	0.0000%	-10.7977%	0.0000%
-207	0.0000%	-6.5793%	0.0000%
-681	0.0000%	-6.0533%	0.0000%
...	...	...	...
-1267	0.0000%	-2.0762%	1.3469%
-842	0.0000%	-2.0643%	1.3469%
-8	3.8909%	-2.0619%	5.2378%
-636	0.0000%	-2.0586%	5.2378%
-242	0.0000%	-2.0492%	5.2378%
...	...	...	...
-733	0.0000%	4.8621%	100.0000%
-698	0.0000%	5.3719%	100.0000%
-575	0.0000%	6.2253%	100.0000%
-734	0.0000%	11.3837%	100.0000%

6. Hybrid VaR: Results

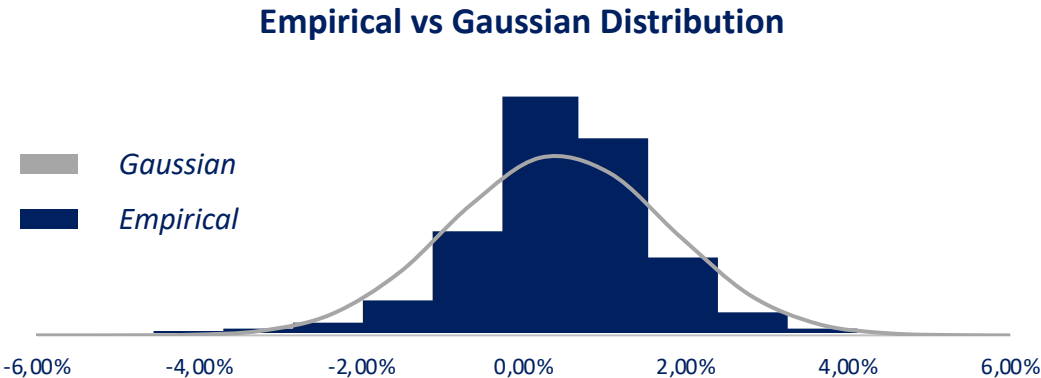
DIFFERENT RISK METRICS

The difference between hybrid and parametric VaR can be explained by the assumption, violated by empirical data, underlying the parametric model.

The VaR provided by the parametric model, with respect to historical data, underestimates extreme negative returns.

For instance, the probability of observing a return equals to -7σ is basically null in a Gaussian distribution, while in the sample we observe it twice.

Type of VaR	Results
Parametric	5.67
Hybrid	6.12
Hybrid (Linear Interpolation)	6.26
Threshold	
Probability in the Gaussian dist.	1.20E-12
Obs. lower than the threshold	2
Frequency in the sample	0.11%



In the hybrid approach no assumptions on the underlying distribution of portfolio returns are made but there is great sampling variability, particularly in the case of tails' events where observations are rarefied.

Unsurprisingly, the Hybrid approach leads to a higher estimate of the VaR, due to the empirical fact that stock returns display the phenomenon of fat tails.

7. LAVaR: From Theory to Reality

A SOLUTION TO UNFORESEEN LOSSES DUE TO MARKET JUMPS CAUSED BY ILLIQUIDITY

LIQUIDITY ADJUSTED VaR: trading losses are not exogenous events → they are phenomena caused by the interaction between the single agent and the market.

- ❑ A trader does not realize the mid-price when liquidating his position, but rather the **mid-price minus half the bid-ask spread and an additional price impact.**
- ❑ *Liquidity risk* associated with the uncertainty of the spread and the additional impact component is particularly important for thinly traded or emerging market securities.
- A way to deal with liquidity risk is to calculate log returns as:

$$R_{i,t} = \ln \left[\frac{P_{i,t} * (1 - S_j * M/2)}{P_{i,t-1}} \right]$$

- S_j is the bid-ask spread of security i at time t
- M is an additional price impact component

MULTIPLE (M)		
Horizon	Value	Criterion
Daily	5	Executing an order over a longer time span dilutes the order's effects on the depth of the trading book
10 days	2	

- μ changes evidently but it is not an input of the VaR (equal to 0 by assumption)

DAILY			
Correlation matrix	IBM	PHG	SHEL
IBM	1	0.22	0.01
PHG	0.22	1	-0.07
SHEL	0.01	-0.07	1

VaR EW Portfolio
→ 5.63

10-DAY			
Correlation matrix	IBM	PHG	SHEL
IBM	1	-0.11	0.31
PHG	-0.11	1	0.08
SHEL	0.31	0.08	1

VaR EW Portfolio
→ 16.32

8. LAVaR: Dealing with Correlations

DEALING WITH CORRELATIONS BETWEEN BID-ASK SPREADS AND BETWEEN MARKET AND LIQUIDITY RISK

CHERUBINI (1997):

$$LAVaR_p = \sqrt{\sum_{i=1}^n \sum_{j=1}^n VaR_i VaR_j \rho(R_i, R_j) + \sum_{i=1}^n \sum_{j=1}^n LA_i LA_j \rho(S_i, S_j) + 2 \sum_{i=1}^n \sum_{j=1}^n VaR_i LA_j \rho(R_i, S_j)} \quad \text{with } LA_j = Size_j * Q_{S_j}(p)/2$$

- ❑ Diversification: in VaR calculations the **spreads S_j** are **additional factors** integrated in the correlation matrix
 - ❑ $Q_{S_j}(p)$ is the spread distribution quantile function and p the probability value at which the LA_j is computed
- We followed Bangia, Diebold, Schuermann and Stroughair (1999) quantile approach to obtain a **worst-case estimate of liquidation costs** with a given degree of confidence → non-parametric estimate of the quantile due to the irregular shape of the empirical distribution of the observed spread (fat tails and bimodality hinting at a regime-switching process of spread innovations) → the normality (Gaussian) assumption may be misleading
 - *10-days empirical spread distributions are the daily observed spreads' arithmetic averages (over 10 days)*
 - **Sub-optimal methodology:** LAVaR is **slightly higher** than the one computed with the parametric approach with a *daily time horizon* and **lower** if comparing the results with a *10-days time horizon*:
 - Mathematically feasible if covariances between spreads and returns are negative, but not economically meaningful: *integrating additional model risks should at least not diminish the overall risk exposure*

9. LAVaR: Time Aggregation

DEALING WITH SERIAL CORRELATION AMONG RETURNS

Due to the additivity of log returns, it would be enticing to compute the variance of the log returns of a given time horizon starting from the variance of the log returns of a given unit of time, but some conditions must be met:

- **In absence of serial correlation:** the variance of the sum equals the sum of the variances: having estimated the variance of the daily log returns, the estimate of the variance of the 10-day log returns is $\hat{\sigma}_{10d}^2 = 10 \hat{\sigma}_d^2$
- **In presence of serial correlation:** the variance of the sum contains covariances between the log returns of different instant of times: to avoid cumbersome estimation of parameters, compute the variance of the 10-days log returns, even if reducing the sample size.

The presence of serial correlation among log returns is verified through the Durbin-Watson test, using the following t statistics:

$$DW = \frac{\sum_{i=1}^n (r_{t-i} - r_{t-i-1})^2}{\sum_{i=2}^n r_{t-i}^2}$$

The DW t statistics is approximately equal to $2(1 - \hat{\rho}_1)$ where $\hat{\rho}_1$ is the one lag autocorrelation function: if there is no autocorrelation, $DW \cong 2$.

The DW t statistics is compared with two critical values, d_L and d_U : (1) if **DW** < **d_L** we reject the null hypothesis (there is **autocorrelation**); (2) if **DW** > **d_u** we accept the null hypothesis.

Durbin-Watson test			
IBM	PHG	SHEL	EW Portfolio
2.00	2.04	1.92	1.96

The null is rejected for SHEL and the portfolio log returns: the variance of the sum is not the sum of the variances

10. LAVaR: Optimal Choice

A CONSERVATIVE AND REALISTIC ESTIMATE

The liquidity component of the VaR can be estimated in several ways, one of which is **not to account for diversification effects** and just adds the liquidity components of each security, leading to a conservative estimate of the Liquidity Adjusted VaR :

$$LAVaR_p = \sqrt{\sum_{i=1}^n \sum_{j=1}^n VaR_i VaR_j \rho(R_i, R_j)} + \sum_{j=1}^n LA_j$$

TWO IMPLICIT ASSUMPTIONS:

- I. The correlation matrix between spreads and returns is a matrix of zeroes
- II. The correlation matrix of the spreads is an identity matrix

The LAVaR measure we select assumes that correlations **across the liquidity risks** of individual stocks and correlations **between spreads and returns** are zero: liquidity and market risks are assumed to be independent. This to avoid a decrease of VaR (caused by negative estimated covariances between spreads and returns) which is difficult to explain from an economic viewpoint.

By construction, the Liquidity Adjusted VaR computed in this way is going to be higher than the one computed with the parametric approach by the sum of the liquidity costs of the securities in the portfolio (the **liquidation costs** are **strictly positive** for every security).

Daily	
Type of VaR	Results
LAVaR - Diversification	5.82
LAVaR - No Diversification	6.31

10 days	
Type of VaR	Results
LAVaR - Diversification	16.08
LAVaR - No Diversification	16.62

11. Conclusions & References

CONCLUSIONS

The **parametric approach** is the one underestimating the VaR since it does not take in consideration fat tails in the return distribution and it does not quantify liquidity risk.

The **hybrid approach** solves the problem of fat tails but suffers from great sampling variability.

The **LAVaR calculated with the multiple approach** is similar to the parametric one since the correlation matrices are pretty similar and setting $\mu=0$ in both of them does not take into consideration that on average «stressed» returns are lower in the LAVaR with multiple approach.

The **LAVaR approach with diversification effects**, despite being the most reasonable on a theoretical level, yields not realistic results, in particular for the 10 days risk horizon, due to the presence of negative correlations between spreads and returns which would lead to negative liquidity costs on the overall portfolio.

The **LAVaR approach with no diversification effects** yields the most prudent and realistic estimate of the VaR, where the liquidity cost is positive and taken in consideration.

Looking at the **difference between the LAVaR measures with different risk horizons** selected (no diversification effects), it can be noticed that the liquidation cost of both the portfolio and the single securities is lower in the 10 days risk horizon:

- The statistical intuition is that the empirical distribution of the 10 days spreads is composed of datapoints which are arithmetic averages: the distribution presents less extreme observations in its left tail, leading to a lower liquidation cost.
- The economic intuition is the one used to justify the multiples.

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Resti, A., & Sironi, A. (2013). *Risk management and shareholders' value in banking: From risk measurement models to capital allocation policies*. John Wiley & Sons.

Daily		10 days	
Type of VaR	Results	Type of VaR	Results
VaR - Parametric	5.67	VaR - Parametric	16.12
LAVaR - Multiple	5.63	LAVaR - Multiple	16.32
LAVaR - Diversification	5.82	LAVaR - Diversification	16.08
LAVaR - No Diversification	6.31	LAVaR - No Diversification	16.62

Daily		10 days	
LA IBM	0.56	LA IBM	0.43
LA PHG	0.09	LA PHG	0.05
LA SHEL	0.06	LA SHEL	0.03
LA EW Portfolio	0.64	LA EW Portfolio	0.51